

ZACHERY PIPES

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EDUCATION

Missouri University of Science and Technology – Rolla, MO

Expected Fall 2027

B.S. in Computer Science

GPA: 3.7 / 4.0

Relevant Coursework: Algorithms, Data Structures, Operating Systems, Databases & File Structures, Computer Networking, Theory of Computer Science, Intro to Parallel Programming, Computer Security, Programming Languages

TECHNICAL SKILLS

Programming Languages: C++, C#, Python, Java

Libraries & APIs: CUDA, pandas, NumPy, requests, eBay, pybind11, OpenCV

Certifications: Java, Microsoft Word, Microsoft Excel, Microsoft PowerPoint

Honors & Awards: S&T Honors Academy, Kummer Vanguard Scholar, Dean's List

PROFESSIONAL EXPERIENCE

Software Team – Special Projects Lead

Aug 2024 – Present

Multirotor Design Team, Missouri S&T

- Directed three engineering sub-teams (9+ members) spanning flight simulation, developer tooling, and 3D terrain mapping
- Led migration of the team simulator from legacy AirSim to Project AirSim, containerizing the Unreal Engine and ArduPilot SITL stack (Podman/WSL2) and restructuring it as a shared submodule serving the design team's competition codebases
- Cut new-member setup from a multi-hour source build to a single pull by publishing prebuilt environment and SITL images to GitHub Container Registry behind a one-command launcher
- Structured sub-team work around end-to-end feature ownership rather than by layer, giving every member hands-on work with the team's core technologies
- Led onboarding for 5+ new members, training them in Python, Git submodule workflows, and containerized development environments

Software Engineer

Apr 2026 – Present

Vortice Avionics – Rolla, MO

- Developed C++ software for a UAV platform, contributing through code review and testing
- Collaborated with 10+ teammates in a student-led UAV startup advancing next-generation UAV technology under active NDA
- Modeled product components in Blender and produced marketing renders used in the company website and pitch decks
- Evaluated candidate UAV components against Blue UAS sourcing requirements, weighing compliance against performance constraints

PROJECTS

fastdist | C++, CUDA, Python, pybind11, NumPy, GitHub Actions

Dec 2025 – Present

- Implemented **26** CUDA kernels accelerating 5 probability distributions and vector utilities, with a streaming executor that splits large inputs across **4** concurrent CUDA streams
- Built the pybind11/NumPy wrapper layer between C++ and Python, releasing the GIL during computation and managing GPU memory through self-freeing buffer objects that are allocated once and reused across calls, replacing manual cleanup on six separate error paths
- Achieved up to **57x** speedup over SciPy with the C++ CPU backend alone, with CUDA kernels adding a further **2.5x** over the CPU path on large inputs
- Automated cross-platform wheel builds for Linux, macOS (x86_64/arm64), and Windows across Python **3.10–3.14**, with trusted-publishing releases to PyPI and a CI check that catches type-stub drift against a CUDA build